# CASE STUDY: DISEASE DIAGNOSIS USING MACHINE LEARNING ALGORITHMS

Machine Learning for Diabetes Prediction

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Course: Machine Learning

# Machine Learning for Diabetes Prediction: A Comparative Analysis with Preprocessing Optimization, Feature Engineering, and Economic Evaluation

## EXECUTIVE SUMMARY

This project develops a machine learning model to predict diabetes using the PIMA Indian Diabetes Dataset. Five algorithms were compared to identify the most suitable model for medical screening applications. The Neural Network model performed best, accurately identifying most diabetic patients in the test set with strong overall performance. Key findings include that median imputation was more effective than mean for this dataset, the combined Glucose × BMI feature emerged as the most important predictor across models, and lowering the decision threshold improved recall, which is important because missing a diabetic patient has more serious consequences than sending a healthy person for follow-up testing. These results indicate that medical ML systems must be carefully validated in terms of preprocessing decisions, domain-informed feature engineering, and configurations aligned with clinical priorities rather than relying solely on statistical metrics.

## PROBLEM DEFINITION

Diabetes is a severe health issue in the world with more than 537 million adults having the disease and it is projected that this number will reach 783 million by 2045. The disease can lead to complications including cardiovascular disease, nerve damage, vision loss, and kidney failure Studies have indicated that about 24 percent of diabetic patients are still not diagnosed, thus complicating the identification of the problem and making them susceptible to the development of complications that can be prevented.

As of now, the diagnosis of the disease is based on blood sampling (measuring a level of fasting glucose of 126mg/dl and above or measuring the level of HbA1c of 6.5 and above). These diagnoses need laboratory findings and trained health professionals which may restrict access under certain places.

The key question that this project aims to answer is the following: Are machine learning models capable of identifying individuals at higher risk of diabetes based on simple measurements, and in that case, how should they be configured for practical use in medical screening? It entails solving the practical data challenges such as missing values and class imbalance, choice of suitable models, and the implications of false negative and false positive in the context of medical screening.

## OBJECTIVES

1. Develop and compare five machine learning models (Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and Neural Network) for predicting diabetes using the PIMA dataset.
2. Evaluate different preprocessing strategies, including handling missing data and addressing class imbalance, to determine their impact on model performance.
3. Investigate the effect of feature engineering by introducing a Glucose × BMI interaction term and analyzing feature importance across models.
4. Assess model performance using multiple evaluation metrics (accuracy, precision, recall, specificity, F1-score, and AUC-ROC), including cross-validation and test set analysis.
5. Examine the effects of hyperparameter tuning and decision threshold adjustment to understand how configuration choices influence real-world medical screening suitability.

## DATASET DESCRIPTION

The PIMA Indian Diabetes Dataset contains medical records of 768 Pima Indian women aged 21 years and older and it was collected by the National Institute of Diabetes and Digestive and Kidney Diseases. There are eight health measures and one outcome variable which is diabetes status.

**Features:**

- **Pregnancies**: Number of times pregnant
- **Glucose**: Blood sugar level (mg/dL; diabetes threshold: 126)
- **BloodPressure**: Diastolic blood pressure (mm Hg)
- **SkinThickness**: Triceps skin fold thickness (mm)
- **Insulin**: Insulin level (μU/mL)
- **BMI**: Body mass index (obesity threshold: 30)
- **DiabetesPedigreeFunction**: Genetic risk score
- **Age**: Age in years
- **Outcome**: Diabetes status (0 = non-diabetic, 1 = diabetic).

**Dataset Issues:**

The data contains class imbalance among 500 non-diabetic cases (65.1 percent) and 268 diabetic cases (34.9 percent). Also, numerous characteristics have physiologically impossible zero values: Insulin (48.7% zero) and SkinThickness (29.6% zero). Such zeros indicate the missing measurements, which require imputation.

## FEATURE ANALYSIS

Exploratory analysis revealed that Glucose was the most influential predictor of diabetes (correlation: 0.467), then there are BMI (0.293), Age (0.238), and Pregnancies (0.222). Statistical tests established that 6 out of eight features have significant differences between diabetic and non-diabetic groups.

Patients with diabetes were found to have a higher level of glucose, BMI and were older than those who did not have diabetes. The data has outliers of extreme values, especially in Insulin (maximum: 846) and SkinThickness resulting in highly skewed distributions. This justifies the use of median imputation instead of mean, as the median is less affected by outliers while the mean is influenced by extreme values.

Certain characteristics are interdependent: SkinThickness and BMI and Insulin (both of these variables are body composition-related), and Age and Pregnancies. Glucose remains relatively independent from other features, making it a distinct predictor.

## DATA PREPROCESSING TECHNIQUES

A six-step preprocessing pipeline was developed, and its major decisions were experimentally validated.

- **Missing Value Imputation**: Three approaches (mean, median, mode) were evaluated by training the same models of the Random Forest. Median imputation achieved the highest AUC-ROC (0.8323) compared to mean (0.8231) and mode (0.8195), outperforming the alternatives by 1.56%. Since the outliers in such features as Insulin are extremely high, the feature of median makes it a better option.
- **Feature Engineering:** Two features have been developed. In the first, Glucose x BMI interaction, based on the observation that elevated blood sugar combined with obesity may indicate metabolic dysfunction. based on the observation that elevated blood sugar combined with obesity may indicate metabolic dysfunction. Second, Age was modeled into groups (21-30, 31-40, 41-50, 51+) but it was not as helpful as continuous age.
- **Data Splitting:** Data were stratified split (80-20, training: 614 samples, testing: 154 samples) and a random seed of 42 was used.
- **Class Balancing:** the study used random oversampling to take care of class imbalance: Diabetic cases were randomly duplicated until they were equally represented (400 samples per class, 800 total training samples).
- **Feature Scaling:** MinMaxScaler scaled all the features between 0-1. The training data was used to compute the scaling parameters to avoid data leakage.
- **Pipeline Integrity:** Pipeline preprocessing was only done on training data, and Test data preprocessing was performed using parameters derived from the training set.

## IMPLEMENTATION OF ML ALGORITHMS

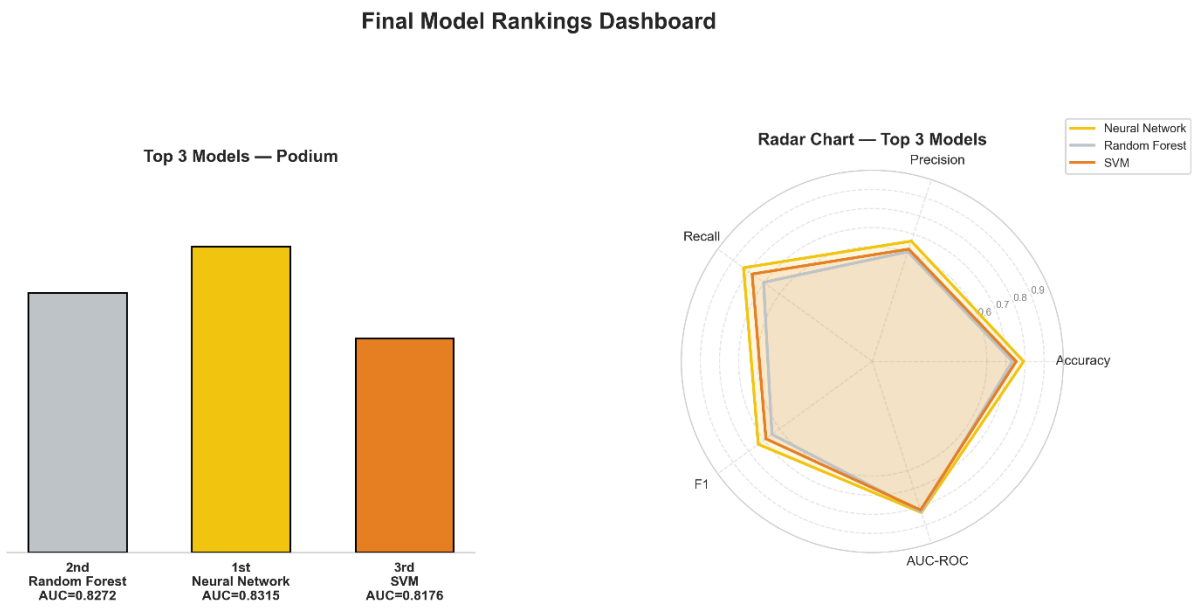
Scikit-learn was used to implement five algorithms:

1. **Logistic Regression:** A linear model that estimates the probability of diabetes using a weighted sum of features. Gives interpretable coefficients which act as a benchmark.
2. **Decision Tree:** Generates rules of the form if-then by splitting up the data recursively. Depth was reduced to 5 with minimal sampling requirements in order to minimize overfitting.
3. **Random Forest:** 100 decision trees with maxdepth of 10 and minsamples of leaf 5. The mean of trees decreases overfitting.
4. **Support Vector Machine:** Determines the best boundary between cases by RBF kernel in the case of non-linear classification. C = 1.0 for moderate regularization.
5. **Neural Network:** Two-layer (64 and 32 neurons) tanh activation and Adam optimizer. Converged in 158 iterations.

## MODEL EVALUATION AND COMPARISON

Models were evaluated using six metrics on 154 test patients (54 diabetic):

| Rank | Model | Accuracy | Precision | Recall | F1-Score | AUC-ROC |
|---|---|---|---|---|---|---|
| **1st** | **Neural Network** | **79.22%** | **66.18%** | **83.33%** | **0.7377** | **0.8315** |
| 2nd | Random Forest | 73.38% | 60.32% | 70.37% | 0.6496 | 0.8272 |
| 3rd | SVM | 75.32% | 61.76% | 77.78% | 0.6885 | 0.8176 |
| 4th | Logistic Regression | 72.73% | 59.68% | 68.52% | 0.6379 | 0.8126 |
| 5th | Decision Tree | 68.83% | 54.69% | 64.81% | 0.5932 | 0.7771 |




**Complete Model Rankings — All Metrics**

| Rank | Model | Accuracy | Precision | Recall | Specificity | F1-Score | AUC-ROC |
|---|---|---|---|---|---|---|---|
| 1st | Neural Network | 79.22% | 66.18% | 83.33% | 77.00% | 73.77% | 0.8315 |
| 2nd | Random Forest | 73.38% | 60.32% | 70.37% | 75.00% | 64.96% | 0.8272 |
| 3rd | SVM | 75.32% | 61.76% | 77.78% | 74.00% | 68.85% | 0.8176 |
| 4th | Logistic Regression | 72.73% | 59.68% | 68.52% | 75.00% | 63.79% | 0.8126 |
| 5th | Decision Tree | 68.83% | 54.69% | 64.81% | 71.00% | 59.32% | 0.7771 |

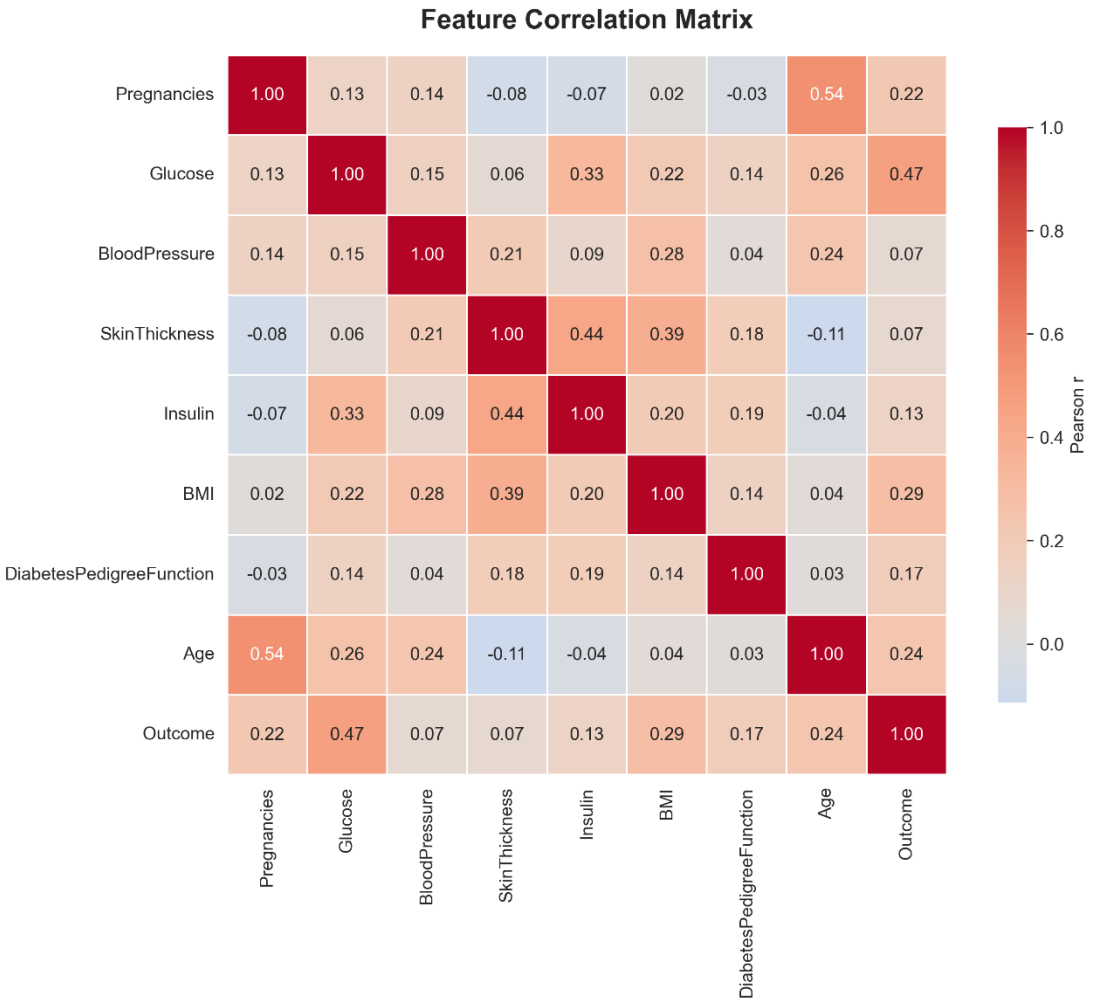
Model performance comparison showing Neural Network achieved highest AUC-ROC (0.8315) and recall (83.33%), missing only 9 of 54 diabetic patients.

The Neural Network achieved an AUC-ROC of 0.8315 and correctly identified 45 of 54 diabetic patients (9 false negatives). The confusion matrix shows 23 false positives and 77 true negatives, yielding 77% specificity. Cross-validation showed acceptable stability (mean accuracy: 79.31% ± 2.23%), indicating consistent performance across different data splits.

## FEATURE IMPORTANCE ANALYSIS

Feature importance was analysed across Logistic Regression, Decision Tree, and Random Forest:

| Rank | Health Feature | Average Importance | Clinical Interpretation |
|---|---|---|---|
| **1st** | **Glucose × BMI Interaction** | **0.986** | **Synergistic risk of hyperglycemia and obesity.** |
| 2nd | Glucose | 0.648 | Primary diagnostic marker for insulin resistance. |
| 3rd | BMI | 0.345 | Indicator of metabolic health and adiposity. |
| 4th | Pregnancies | 0.251 | History of gestational metabolic stress. |
| 5th | Diabetes Pedigree Function | 0.237 | Genetic predisposition based on family history. |



Feature importance across three models. The engineered Glucose×BMI interaction ranked first with importance of 0.986.

The engineered Glucose×BMI interaction feature showed the highest importance (0.986), ranking first across both tree-based models and near-maximum for Logistic Regression. This exceeds individual Glucose and BMI importance, suggesting the multiplicative combination captures valuable patterns.

All models agree on the importance of Glucose and BMI. However, disagreements appear for Age: Logistic Regression assigns minimal importance (0.018) while Random Forest assigns moderate importance (0.360), suggesting non-linear age effects that tree models capture but linear models cannot.

## HYPERPARAMETER TUNING STUDY

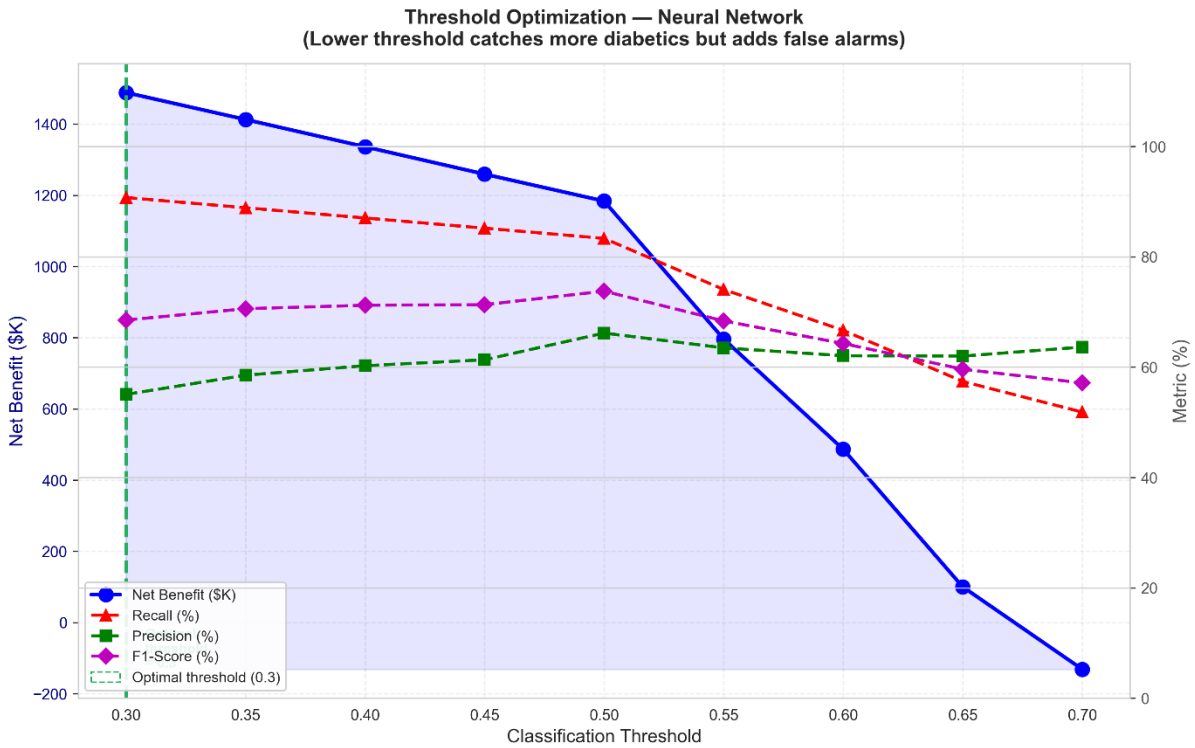
GridSearchCV was conducted using 96 parameter combinations. The hyperparameter tuning for Random Forest evaluated parameters including n_estimators, max_depth, min_samples_leaf, and max_features. The cross-validation AUC also increased (0.8272 to 0.9316) but test recall went down (70.37 to 62.96). This indicates potential overfitting, in which the minsamplesleaf parameter of 1 permitted trees to remember training patterns. The default setting of minsamples leaf of 5 gave increased regularization. This suggests that cross-validation results alone may be misleading, and optimizing one metric can negatively affect other clinically important measures.

## THRESHOLD OPTIMIZATION FOR CLINICAL DEPLOYMENT

The default classification thresholds are usually 0.5, yet medical screening uses can find alternative values useful as the consequences of false negative (diagnoses not performed) are more severe compared to false positive (needless follow-ups).

Testing different thresholds on the Neural Network revealed:

| Classification Threshold | True Positives (TP) | False Negatives (FN) | Recall (Sensitivity) | Clinical Impact |
|---|---|---|---|---|
| **0.30 (Optimal)** | **49** | **5** | **90.74%** | **Highest detection; lowest risk of missed cases.** |
| 0.50 (Default) | 45 | 9 | 83.33% | Balanced approach; moderate risk of missed cases. |
| 0.70 (Conservative) | 37 | 17 | 68.52% | High specificity; high risk of missing sick patients. |




Threshold analysis showing lowering threshold from 0.5 to 0.3 improves recall from 83.33% to 90.74%, detecting 4 additional diabetic patients.

Lowering the threshold to 0.30 detects 4 more diabetic patients while generating 8 additional false positives. For preliminary screening where early detection enables intervention and false positives lead only to confirmatory testing, this trade-off may be reasonable. The specific

threshold would require clinical validation before deployment, as optimal values depend on population demographics, testing availability, and healthcare capacity.

## RESULT INTERPRETATION

Therefore, the default parameters were retained.The Neural Network achieved the best overall performance, with an AUC-ROC of 0.8315 and 83.33% recall, missing 9 of 54 diabetic patients. The cross-validation indicated good stability and as such the model can be expected to apply to new patients.

The major findings include: median imputation model worked better than mean with this dataset (1.56% better), the Glucose x BMI interaction feature ranked the highest among all the models, hyperparameter tuning reduced the test performance yet the cross-validation scores were higher, and the threshold adjustment to 0.30 raised the recall to 90.74%.

The findings indicate that medical ML development needs to be empirically validated on preprocessing decisions, feature engineered by the domain requirements, cross-validation-independent evaluation, and clinical priority configuration settings.

## CONCLUSIONS

This project developed a diabetes prediction model achieving 83.33% recall using a Neural Network. The proposed model is a featured two-layer Neural Network with tanh activation, median imputation, feature creation (using Glucose x BMI), random oversampling, and MinMaxScaler normalization. There could be a more appropriate threshold of 0.30 rather than 0.50 to be used in the screening contexts.

The model may become a risk assessment tool in the initial stages of patient identification to guide confirmatory testing and enable more efficient use of diagnostic resources. When deployed, the system would need continuous observation of recall, false negative rate and AUC-ROC with retraining being done when performance drops.

Such additional steps would be required before deployment: testing on external data, particularly broader an external population than Pima Indian women, integration with electronic health record systems, clinical verification of threshold decisions, and research on explainability techniques.

This piece of work shows that the development of medical ML can entail a proper data preparation, domain-sensitive features creation, multi-metric analysis, and configuration options that are suitable in the target scenario.

## BENEFITS

The model has the potential to enable initial risk assessment using easily accessible measurements. Its 83.33% recall rate increases to 90.74% at adjusted thresholds and is likely to detect the majority of diabetic cases, but could be applicable in the setting with no immediate lab access available. The system delivers the evaluation which is consistent and as close to instant predictions as possible. The preprocessing validation strategy has rigorous method choice, feature engineering is an illustration that medical insight may be utilized to provide direction in design, and the threshold framework might be adjusted to various clinical conditions.

## LIMITATIONS

Data Constraints: The PIMA dataset includes only women Pima Indian women aged 21 years and above, and the findings cannot be applied to other populations. The sample size of 768 patients is relatively small. Approximately half of the Insulin values are missing and SkinThickness values are missing in 29.6%. The dataset dates from the 1990s.

Model Constraints: The modeling can only predict actual diabetes status and not the onset, severity, and type of diabetes. The Neural Network does not have specific prediction interpretation. Predictions do not have confidence intervals. Progression cannot be forecasted using cross-sectional data.

Threshold Considerations: The proposed threshold of 0.30 will require clinical testing. Ideal thresholds depend on the local conditions of diabetes, availability of tests, healthcare capacities, and preferences of the patients.

## ETHICAL CONSIDERATIONS

**Fairness:** Because the model was trained only on Pima Indian women, performance may differ in other populations. Implementation requires demographic-stratified evaluation and perhaps population-stratified to prevent rolling out disproportionality.

**Misclassification Impact:** The missing 5-9 diabetic patients are people that will not get immediate treatment. Every instance of a false positive is subject to unneeded testing which generates anxiety and expenditures. Clear communication is required to ensure the model supports, rather than replaces, clinical judgment.

**Privacy:** This has to be enforced with a high guarantee of data security (encryption, access controls), as well as adherence to privacy policies. Making informed consent to automated assessment should be done by patients.

**Interpretability:** The Neural Network is incapable of explaining such predictions individually and this may decrease trust. Greater interpretability could be beneficial even where less accurate models are used to make high-stake decisions.

**Appropriate Use:** Health care givers should not be too dependent on predictions. Training must be focused on the fact that the system supports, not replaces the medical expertise.

**Deployment Requirements:** To be used responsibly, a clear communication regarding limitations, testing on a wide variety of populations, placement as preliminary screening, intense security infrastructure, continuous monitoring, and human control of the decisions are required.

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